Title: LED Brightness Control using a Potentiometer(Arduino Uno)

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AIM: The aim of this project was to make a control system that could control the brightness of an LED using a potentiometer and Arduino Uno.

COMPONENTS:

1. Potentiometer
2. Arduino Uno
3. LED
4. 220 Ω resistor
5. Jumper wires
6. Breadboard

The Project:

This project demonstrates the use of a potentiometer and an Arduino UNO to control the brightness of LED. The project works on the principle of Pulse Width Modulation(PWM).

Potentiometer: The potentiometer is a variable resistor. It allows us to change it by rotating the knob. It acts as a device to control the LED brightness by varying the resistance of it. Varying the resistance of the Pot changes the voltage on the middle terminal. The potentiometer has 5V across the track and when we rotate the knob, the wiper gives 0-5V on the middle pin according to it's position.

Arduino UNO: This board acts as the brain of the circuit. It controls the brightness by reading the potentiometer's input, processing the value and then generating the PWM signal for the LED.

LED: This is the output device for the circuit. It's a component that gives off light and works only in one direction. Since a LED has no resistance of it's own without an external resistor the LED would burn out.

Resistor: This is the component that limits the current through the LED. It sits in series with the LED.

Breadboard: The pot's three pins and the LED go through the breadboard.

Jumper Wires: They connect all the components together.

Working:

This project works on the principle of Pulse Width Modulation(PWM).

The UNO runs on 5V logic level. It uses PWM for outputting an intermediate voltage because it cannot output a variable voltage. The pin can only be at a HIGH(5V) or a LOW(0V). So PWM makes the pin switch ON and OFF really quickly so that the average output voltage is intermediate. The higher the duty cycle, the higher the average power delivered is higher.

The time for which the pin stays at HIGH during 1 ON/OFF cycle is known as the Duty Cycle. Only pins 3,5,6,9,10 & 11 can support PWM on the UNO.

The potentiometer is connected between the 5V and the GND pins of the board and the middle pin is connected to the A0 pin. When the knob of the potentiometer is rotated, the resistance ratio changes and a variable voltage is produced on the middle pin. This voltage is then converted by the A0 pin to a digital value 0-1023. This is then converted again into a value between 0-255 for the PWM as it only uses 8 bit resolution.

That value determines the brightness of the LED. PWM controls the LED.

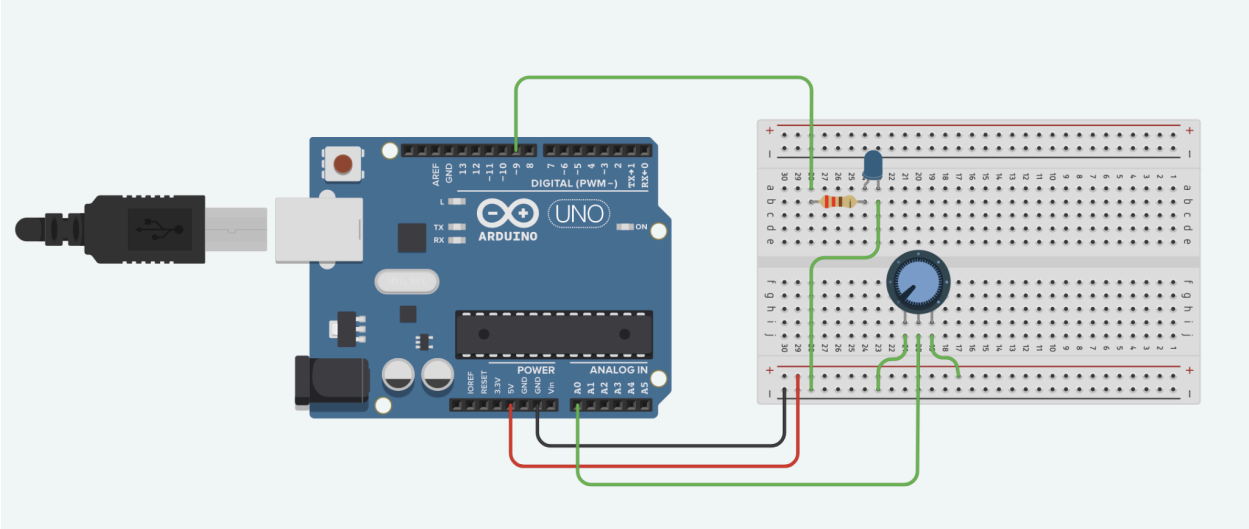
Circuit:

The potentiometer's left terminal is connected to the 5V of the arduino UNO.
The right terminal is connected to the GND of the UNO.
The middle pin is connected to the A0.
The LED's anode is connected to the ~9 digital pin on the board through a 220 Ω resistor and the cathode is connected to the GND.

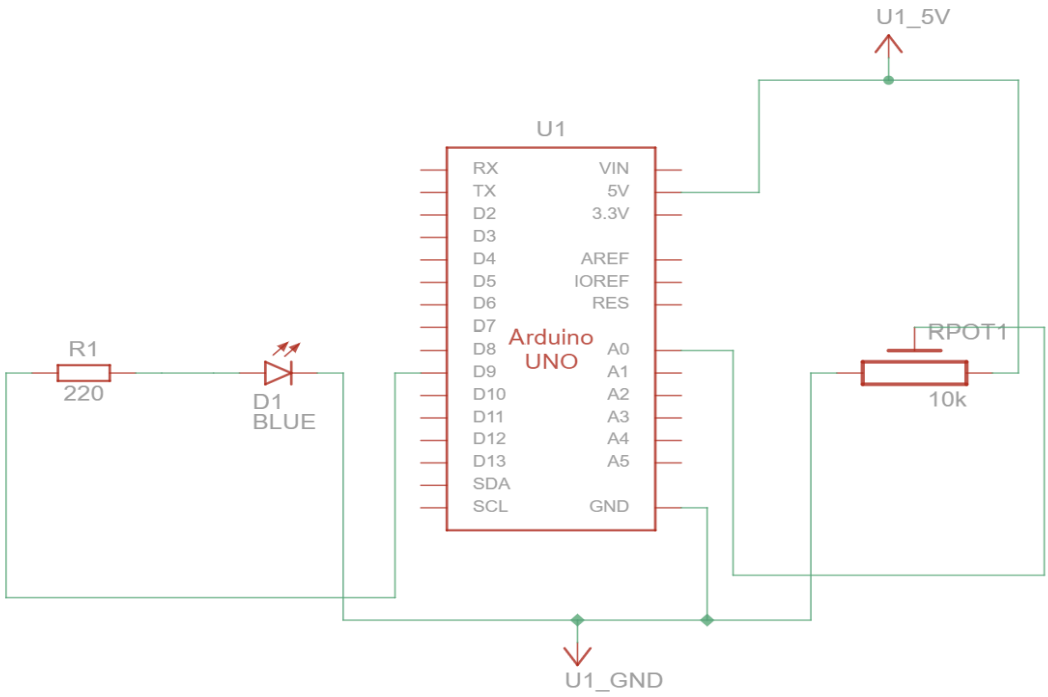
Component	Component Pin	Arduino Pin	Purpose
Potentiometr	Left Terminal	**GND**	Ground Reference
	Middle Wiper	**A0**	Analog Input Signal
	Right Terminal	**5V**	5V Power Reference
Blue LED	Anode (+)	**D9**	PWM Variable Output
	Cathode (-)	**GND**	Ground Return

Resistor	Inline	**D9**	LED Protection (220Ω)

Breadboard Layout:



Circuit Schematic:



Project Code:

```
// Potentiometer-controlled LED brightness using PWM

int potPin = A0;
int ledPin = 9;

int potValue = 0;
int brightness = 0;

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);  // Initializes serial communication at 9600 baud
}

void loop() {
  potValue = analogRead(potPin);  // Reads potentiometer value (0-1023)

  brightness = map(potValue, 0, 1023, 0, 255);  // Converts reading to PWM
range (0-255)

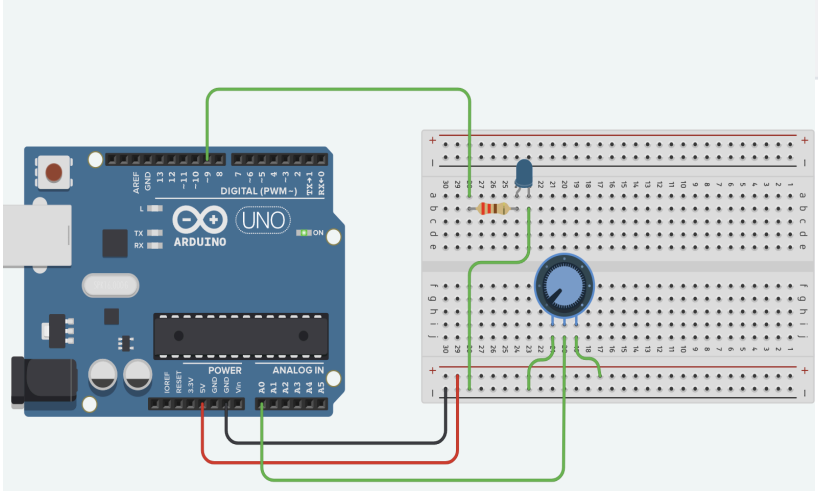
  analogWrite(ledPin, brightness);  // Controls LED brightness using PWM

  Serial.print("Potentiometer Value: ");
  Serial.print(potValue);  // Displays analog reading

  Serial.print(" | LED Brightness: ");
  Serial.println(brightness);  // Displays PWM value and moves to next line

  delay(100);  // Waits 100 milliseconds before next reading
}
```

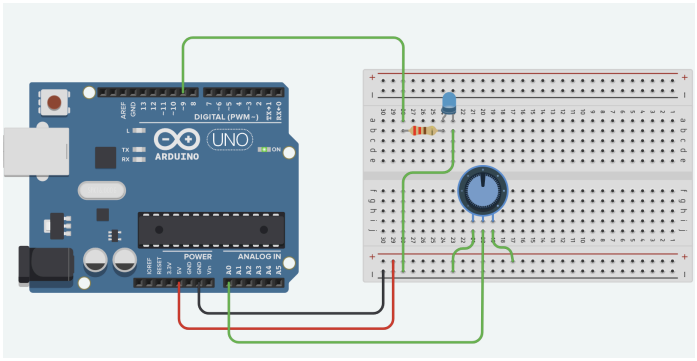
Project Outputs:



Serial Monitor

```
Potentiometer Value: 0 | LED Brightness: 0
Potentiometer Value: 0 | LED Brightness: 0
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```

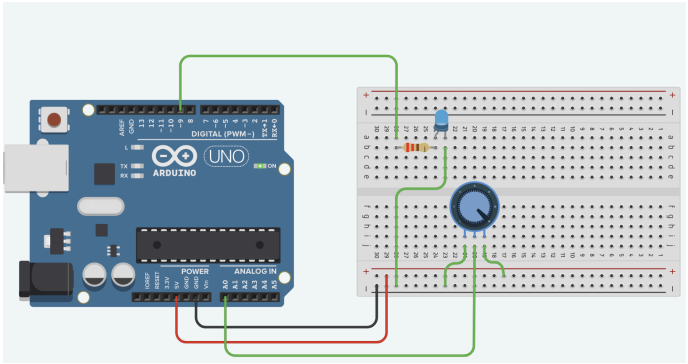
1.Output in the serial monitor when the potentiometer is not rotated.



Serial Monitor

```
Potentiometer Value: 511 | LED Brightness: 127
Potentiometer Value: 511 | LED Brightness: 127
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Potentiometer Value: 511 | LED Brightness: 127
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Potentiometer Value: 511 | LED Brightness: 127
Potentiometer Value: 511 | LED Brightness: 127
```

2. Output in serial monitor when the pot is rotated 50% (512)



Serial Monitor

```
Potentiometer Value: 1023 | LED Brightness: 255
Potentiometer Value: 1023 | LED Brightness: 255
Potentiometer Value: 1023 | LED Brightness: 255
Potentiometer Value: 1023 | LED Brightness: 255
Potentiometer Value: 1023 | LED Brightness: 255
Potentiometer Value: 1023 | LED Brightness: 255
Potentiometer Value: 1023 | LED Brightness: 255
```

3. Output in serial monitor when the pot is rotated 100%(1023)

Project link on Tinkercad:

POT LED DIMMER

Conclusion:

This project varies the brightness of an LED smoothly using a potentiometer and Arduino UNO through PWM. This can be varied smoothly from OFF to maximum brightness due to the effective PWM based controlling.